

Original LCTRSs

LCTRS

1

$$\Sigma = \Sigma_{theory}^{int} \cup \left\{ \begin{array}{l} \text{config} : \text{Thread} \times \text{Thread} \times \text{Thread} \times \text{Thread} \times \text{Thread} \times \text{int} \times \text{int} \Rightarrow \text{Config} \\ \text{thread} : \text{State} \Rightarrow \text{Thread} \\ \text{none} : \text{Thread} \\ \text{main} : \text{State} \\ \text{main0} : \text{State} \\ \text{main1} : \text{State} \\ \text{main2} : \text{State} \\ \text{main3} : \text{State} \\ \text{main4} : \text{State} \\ \text{main5} : \text{State} \\ \text{main6} : \text{State} \\ \text{main7} : \text{State} \\ \text{return0} : \text{State} \\ \text{setThread} : \text{State} \\ \text{setThread1} : \text{State} \\ \text{setThread2} : \text{State} \\ \text{checkThread} : \text{State} \\ \text{success} : \text{Config} \\ \text{error} : \text{Config} \\ \text{assert0} : \text{Config} \end{array} \right\}$$

2

1 :	config(thread(main), none, thrd1, thrd2, thrd3, a, b) → config(thread(main0), thread(setThread), thrd1, thrd2, thrd3, a, b)	
2 :	config(thread(main), thread(s), thrd1, thrd2, thrd3, a, b) → error	
3 :	config(thread(main0), thrd0, none, thrd2, thrd3, a, b) → config(thread(main1), thrd0, thread(setThread), thrd2, thrd3, a, b)	
4 :	config(thread(main0), thrd0, thread(s), thrd2, thrd3, a, b) → error	
5 :	config(thread(main1), thrd0, thrd1, none, thrd3, a, b) → config(thread(main2), thrd0, thrd1, thread(setThread), thrd3, a, b)	
6 :	config(thread(main1), thrd0, thrd1, thread(s), thrd3, a, b) → error	
7 :	config(thread(main2), thrd0, thrd1, thrd2, none, a, b) → config(thread(main3), thrd0, thrd1, thrd2, thread(checkThread), a, b)	
8 :	config(thread(main2), thrd0, thrd1, thrd2, thread(s), a, b) → error	
9 :	config(thread(main3), thread(return0), thrd1, thrd2, thrd3, a, b) → config(thread(main4), none, thrd1, thrd2, thrd3, a, b)	
10 :	config(thread(main4), thrd0, thread(return0), thrd2, thrd3, a, b) → config(thread(main5), thrd0, none, thrd2, thrd3, a, b)	
11 :	config(thread(main5), thrd0, thrd1, thread(return0), thrd3, a, b) → config(thread(main6), thrd0, thrd1, none, thrd3, a, b)	
12 :	config(thread(main6), thrd0, thrd1, thrd2, thread(return0), a, b) → config(thread(main7), thrd0, thrd1, thrd2, none, a, b)	
13 :	config(thread(main7), thrd0, thrd1, thrd2, thrd3, a, b) → config(thread(return0), thrd0, thrd1, thrd2, thrd3, a, b)	
14 :	config(thrd, thread(setThread), thrd1, thrd2, thrd3, a, b) → config(thrd, thread(setThread1), thrd1, thrd2, thrd3, a1, b)	[a1 = 1]
15 :	config(thrd, thread(setThread1), thrd1, thrd2, thrd3, a, b) → config(thrd, thread(setThread2), thrd1, thrd2, thrd3, a, b1)	[b1 = -1]
16 :	config(thrd, thread(setThread2), thrd1, thrd2, thrd3, a, b) → config(thrd, thread(return0), thrd1, thrd2, thrd3, a, b)	
17 :	config(thrd, thrd0, thread(setThread), thrd2, thrd3, a, b) → config(thrd, thrd0, thread(setThread1), thrd2, thrd3, a1, b)	[a1 = 1]
18 :	config(thrd, thrd0, thread(setThread1), thrd2, thrd3, a, b) → config(thrd, thrd0, thread(setThread2), thrd2, thrd3, a, b1)	[b1 = -1]
19 :	config(thrd, thrd0, thread(setThread2), thrd2, thrd3, a, b) → config(thrd, thrd0, thread(return0), thrd2, thrd3, a, b)	
20 :	config(thrd, thrd0, thrd1, thread(setThread), thrd3, a, b) → config(thrd, thrd0, thrd1, thread(setThread1), thrd3, a1, b)	[a1 = 1]
21 :	config(thrd, thrd0, thrd1, thread(setThread1), thrd3, a, b) → config(thrd, thrd0, thrd1, thread(setThread2), thrd3, a, b1)	[b1 = -1]
22 :	config(thrd, thrd0, thrd1, thread(setThread2), thrd3, a, b) → config(thrd, thrd0, thrd1, thread(return0), thrd3, a, b)	
23 :	config(thrd, thrd0, thrd1, thrd2, thread(checkThread), a, b) → assert0	[¬(((a = 0 ∧ b = 0) ∨ (a = 1 ∧ b = -1)))]
24 :	config(thrd, thrd0, thrd1, thrd2, thread(checkThread), a, b) → config(thrd, thrd0, thrd1, thrd2, thread(return0), a, b)	[(a = 0 ∧ b = 0) ∨ (a = 1 ∧ b = -1)]
25 :	config(thrd, thrd0, thrd1, thrd2, thrd3, a, b) → success	

APR Problem

{ 26 : config(thread(main), none, none, none, none, 0, 0) ⇒ success }

BV-LCTRS Version

LCTRS

$$\Sigma = \Sigma_{theory}^{int} \cup \left\{ \begin{array}{l} \text{config} : \text{Thread} \times \text{Thread} \times \text{Thread} \times \text{Thread} \times \text{Thread} \times \text{bitvec}_{32} \times \text{bitvec}_{32} \Rightarrow \text{Config} \\ \text{thread} : \text{State} \Rightarrow \text{Thread} \\ \text{none} : \text{Thread} \\ \text{main} : \text{State} \\ \text{main0} : \text{State} \\ \text{main1} : \text{State} \\ \text{main2} : \text{State} \\ \text{main3} : \text{State} \\ \text{main4} : \text{State} \\ \text{main5} : \text{State} \\ \text{main6} : \text{State} \\ \text{main7} : \text{State} \\ \text{return0} : \text{State} \\ \text{setThread} : \text{State} \\ \text{setThread1} : \text{State} \\ \text{setThread2} : \text{State} \\ \text{checkThread} : \text{State} \\ \text{success} : \text{Config} \\ \text{error} : \text{Config} \\ \text{assert0} : \text{Config} \end{array} \right\}$$

4 {

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1 :      config(thread(main), none, thrd1, thrd2, thrd3, a, b) → config(thread(main0), thread(setThread), thrd1, thrd2, thrd3, a, b)
2 :      config(thread(main), thread(s), thrd1, thrd2, thrd3, a, b) → error
3 :      config(thread(main0), thrd0, none, thrd2, thrd3, a, b) → config(thread(main1), thrd0, thread(setThread), thrd2, thrd3, a, b)
4 :      config(thread(main0), thrd0, thread(s), thrd2, thrd3, a, b) → error
5 :      config(thread(main1), thrd0, thrd1, none, thrd3, a, b) → config(thread(main2), thrd0, thrd1, thread(setThread), thrd3, a, b)
6 :      config(thread(main1), thrd0, thrd1, thread(s), thrd3, a, b) → error
7 :      config(thread(main2), thrd0, thrd1, thrd2, none, a, b) → config(thread(main3), thrd0, thrd1, thrd2, thread(checkThread), a, b)
8 :      config(thread(main2), thrd0, thrd1, thrd2, thread(s), a, b) → error
9 :      config(thread(main3), thread(return0), thrd1, thrd2, thrd3, a, b) → config(thread(main4), none, thrd1, thrd2, thrd3, a, b)
10 :     config(thread(main4), thrd0, thread(return0), thrd2, thrd3, a, b) → config(thread(main5), thrd0, none, thrd2, thrd3, a, b)
11 :     config(thread(main5), thrd0, thrd1, thread(return0), thrd3, a, b) → config(thread(main6), thrd0, thrd1, none, thrd3, a, b)
12 :     config(thread(main6), thrd0, thrd1, thrd2, thread(return0), a, b) → config(thread(main7), thrd0, thrd1, thrd2, none, a, b)
13 :     config(thread(main7), thrd0, thrd1, thrd2, thrd3, a, b) → config(thread(return0), thrd0, thrd1, thrd2, thrd3, a, b)
14 :     config(thrd, thread(setThread), thrd1, thrd2, thrd3, a, b) → config(thrd, thread(setThread1), thrd1, thrd2, thrd3, a1, b) [ a1 = #x00000001 ]
15 :     config(thrd, thread(setThread1), thrd1, thrd2, thrd3, a, b) → config(thrd, thread(setThread2), thrd1, thrd2, thrd3, a, b1) [ b1 = #x11111111 ]
16 :     config(thrd, thread(setThread2), thrd1, thrd2, thrd3, a, b) → config(thrd, thread(return0), thrd1, thrd2, thrd3, a, b)
17 :     config(thrd, thrd0, thread(setThread), thrd2, thrd3, a, b) → config(thrd, thrd0, thread(setThread1), thrd2, thrd3, a1, b) [ a1 = #x00000001 ]
18 :     config(thrd, thrd0, thread(setThread1), thrd2, thrd3, a, b) → config(thrd, thrd0, thread(setThread2), thrd2, thrd3, a, b1) [ b1 = #x11111111 ]
19 :     config(thrd, thrd0, thread(setThread2), thrd2, thrd3, a, b) → config(thrd, thrd0, thread(return0), thrd2, thrd3, a, b)
20 :     config(thrd, thrd0, thrd1, thread(setThread), thrd3, a, b) → config(thrd, thrd0, thrd1, thread(setThread1), thrd3, a1, b) [ a1 = #x00000001 ]
21 :     config(thrd, thrd0, thrd1, thread(setThread1), thrd3, a, b) → config(thrd, thrd0, thrd1, thread(setThread2), thrd3, a, b1) [ b1 = #x11111111 ]
22 :     config(thrd, thrd0, thrd1, thread(setThread2), thrd3, a, b) → config(thrd, thrd0, thrd1, thread(return0), thrd3, a, b)
23 :     config(thrd, thrd0, thrd1, thrd2, thread(checkThread), a, b) → assert0 [ ¬(((a = #x00000000 ∧ b = #x00000000) ∨ (a = #x00000001 ∧ b = #x11111111)) ∧ b = #x00000000) ∨ (a = #x00000001 ∧ b = #x11111111) ]
24 :     config(thrd, thrd0, thrd1, thrd2, thread(checkThread), a, b) → config(thrd, thrd0, thrd1, thrd2, thread(return0), a, b)
25 :     config(thrd, thrd0, thrd1, thrd2, thrd3, a, b) → success

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APR Problem

{ 26 : config(thread(main), none, none, none, none, #x00000000, #x00000000) ⇒ success }